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# Blue Team Handbook Incident Response



Don Murdoch

## **Praise for *Blue Team Handbook: Incident Response***

*Don Murdoch has created an indispensable field guide that belongs on every incident responder's desk. It's packed with practical commands, real-world techniques, and hard-won wisdom from decades in the trenches.*

—Justin Henderson, CEO of Tellaro, Inc.

*Don Murdoch takes his many years of experience and boils them down into a handbook that will help you be a better blue teamer!*

—Doug Burks, founder and CEO of Security Onion Solutions

*Most incident response books give you theory or they give you commands. This one does both. It walks you through building an actual incident response program, the metrics, the leadership structure, the stuff that keeps you from flailing when things go sideways, and then hands you the PowerShell and packet captures to execute.*

—James “@whiskeyhacker” McMurry, CEO, ThreatHunter.ai

*Don's coverage of incident response kept me reading through the night. Anyone in IR should keep this book in their go-bag.*

—Dean Bushmiller, president, Expanding Security

*Don Murdoch connects incident response to real adversary behavior in a way that is immediately useful, tying investigation steps to frameworks like MITRE ATT&CK and practical blue team decision-making. This is a handbook defenders can use to think clearly, prioritize quickly, and improve detection and response with purpose.*

—Tannu Jiwnani, principal security engineer

*If you defend systems for a living, this book belongs on your desk. Don Murdoch cuts through the noise and delivers an incident response playbook that's practical, current, and battle-tested.*

—Sri Sai Bhargav Tiruveedhula, principal security engineer at Autodesk

*An essential reference for blue teams at any stage. Deeply practical, and sharply aligned with today's adversary landscape.*

—Yaamini Barathi Mohan, award-winning cybersecurity leader

*Don is one of the brightest cyber professionals that I've had the pleasure to work with. His skills and knowledge are deep in incident response, endpoint security, and the controls that defend critical IT infrastructure. This book is great for any blue team cyber professional!*

—Alex Kahn, staff cybersecurity engineer, Guidewire Software

*Few books manage to be both a day-one field manual and a long-term desk reference; this one succeeds at both.*

—Nikhil Teja Dommeti, product innovator and security researcher at a Fortune 100 company

*As someone who has been mentored by Don, I've seen firsthand his dedication to the craft of cybersecurity. This updated edition is like having a master teacher by your side, distilling complex concepts into the actionable, real-life examples that define our field.*

— Jack Callaway, PhD., SOC manager

# Blue Team Handbook: Incident Response

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## **Blue Team Handbook: Incident Response**

by Don Murdoch

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## Preface

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A lot has changed in the years since the first edition of *Blue Team Handbook: Incident Response (BTHb:INRE)* was published in 2014. A few blue team-focused books have appeared; meanwhile, red team books still proliferate. If you are not familiar with these terms as they relate to cybersecurity, the blue team is charged with monitoring and defending the organization's information systems—from endpoints to on-premises servers to cloud service providers to software as a service (SaaS)—and with ensuring that the endpoint uses all these resources securely. The red team has the task of behaving like an adversary, which involves external penetration testing and taking advantage of any successes to see how far it can get into the network, explore, and find valuable resources.

Adversaries are effectively pummeling modern businesses and governments into digital submission, and as a result, *BTHb:INRE* must evolve. This edition features numerous updates. These include enhancements to the Preparation, Identification, Containment, Eradication, Recovery, and Lessons Learned (PICERL) incident response (IR) process; memory analysis quick steps; automation tools; Linux lsof; a Volatility 3 quick reference; and a new chapter on PowerShell for Windows. Many of the techniques described can be used in all phases of the IR process. It is my sincere desire that you will continue to gain value from the *BTHb* series and that this book will prove to be a reference you keep handy.

## Who Should Read This Book

Incident response is a critical aspect of security operations and follows a well-structured process. Incident responders often find themselves needing key pieces of information or thought processes to safely move on to the next phase, all the while realizing that continued discovery can cause rescoping

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of the incident. *Blue Team Handbook* provides responders with immediately applicable techniques to handle security incidents today and is filled with life lessons learned from the field. Whether you are new to the field, work in a security operations center and want to move up to the next level, or are a seasoned pro, there is something here for you to up your game.

## Why I Wrote This Book

The first edition of *BTHb:INRE* focused on very practical advice and notes from the field to aid anyone getting into the IR part of the cybersecurity profession. It was intentionally terse. Today, with ever-capable adversaries who are leveraging all kinds of AI and LLM tools, responders have to grow, establish new skills, and take every aspect of the IR game to the next level. I wrote this edition to introduce new skills, thought patterns, capabilities, and lifecycles that improve and update all aspects of the IR process.

## Navigating This Book

**Chapter 1** provides several working definitions of incident response and then covers two aspects of how IR has changed to respond to today's threat actors. Next, it presents several topics that help define some of the key thought processes for IR with time-based security: a discussion of how to leverage the MITRE ATT&CK framework (<https://attack.mitre.org>) and an outline of how adversaries operate today.

**Chapter 2** presents the overall recipe for the IR process. The chapter offers a checklist for all the phases, from Preparation to Lessons Learned, to help you apply a structure and framework to your IR processes.

**Chapter 3** covers a wide variety of skills and tactics that augment the entire IR process, ranging from roles to templates to the traffic light protocol for information sharing.

**Chapter 4** provides an overview of adversary tactics, tools, and procedures, most of which can be encountered during an actual incident.

Chapters **5**, **6**, and **7** go over specific technical aspects of the IR process and examine various platforms.

**Chapter 5** covers Windows examination using a variety of command-line tools; **Chapter 6** focuses on triage and collecting volatile data on Linux systems; and **Chapter 7** discusses PowerShell.

**Chapter 8** provides information on analyzing Active Directory.

**Chapter 9** covers network examinations at the packet capture and traffic levels.

**Chapter 10** discusses Endpoint Detection and Response capabilities.

The book also includes appendices with reference materials, including a list of common TCP and UDP ports.

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*BTHb:INRE* has a companion [GitHub repository and wiki](#), providing numerous scripts and command-line analysis techniques drawn from my experience, many of which are used in this book.

## Conventions Used in This Book

The following typographical conventions are used in this book:

### *Italic*

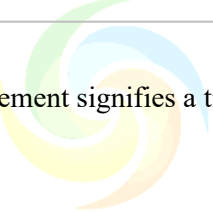
Indicates new terms, URLs, email addresses, filenames, and file extensions.

### `Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

### **Constant width bold**

Shows commands or other text that should be typed literally by the user.



**TIP**

This element signifies a tip or suggestion.

**NOTE**

This element signifies a general note.

ASPOSE  
Your File Format APIs

**WARNING**

This element indicates a warning or caution.

## Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at <https://github.com/DonMVB/BlueTeam-Handbook>.

If you have a technical question or a problem using the code examples, please send an email to [support@oreilly.com](mailto:support@oreilly.com).

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## Chapter 1. Practical Incident Response Defined

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There are two key definitions of *incident response* (IR) that most organizations adopt when building an IR program and its supporting processes: those from NIST and SANS.

From *NIST*

The remediation or mitigation of violations of security policies and recommended practices

From *SANS*

The structured process of identifying, managing, and mitigating the effects of cybersecurity incidents to minimize damage, recover operations, and prevent future occurrences

*BTHb* provides a working definition of IR as the technical aspects of preparing for and responding to an information system–related security incident, including managing the incident through a defined lifecycle.

Regardless of which IR definition you choose, the *BTHb* definition of a *cybersecurity incident* is an adverse condition within the enterprise and/or the systems supporting its mission and objectives. It can be argued that this definition has an overreaching scope because it can easily include events outside the

organization's control. Today, we operate in an interconnected world and may be exposed to another organization's incident. For example, if key suppliers suffer a ransomware attack or a prolonged distributed-denial-of-service (DDoS) attack, our organization can be affected.

Now that IR is defined, the next step is to understand the IR lifecycle. Several frameworks have emerged over the past 30 years to guide security operations in improving their IR processes.

The following topics will be discussed in this chapter:

- The **NIST IR lifecycle**, which is the de facto US federal standard for incident response
- The **SANS PICERL model**
- Important ways to measure protection, detection, and response times as outlined by Winn Schwartau in his book *Time Based Security*
- The **MITRE ATT&CK™ Framework**

For further reading on IR, consider:

- **ISO/IEC 27035**, Information Security Incident Management
- **ENISA's "Good Practice Guide for Incident Management"**
- The Australian Signals Directorate's (ASD) "Strategies to Mitigate Cyber Security Incidents."

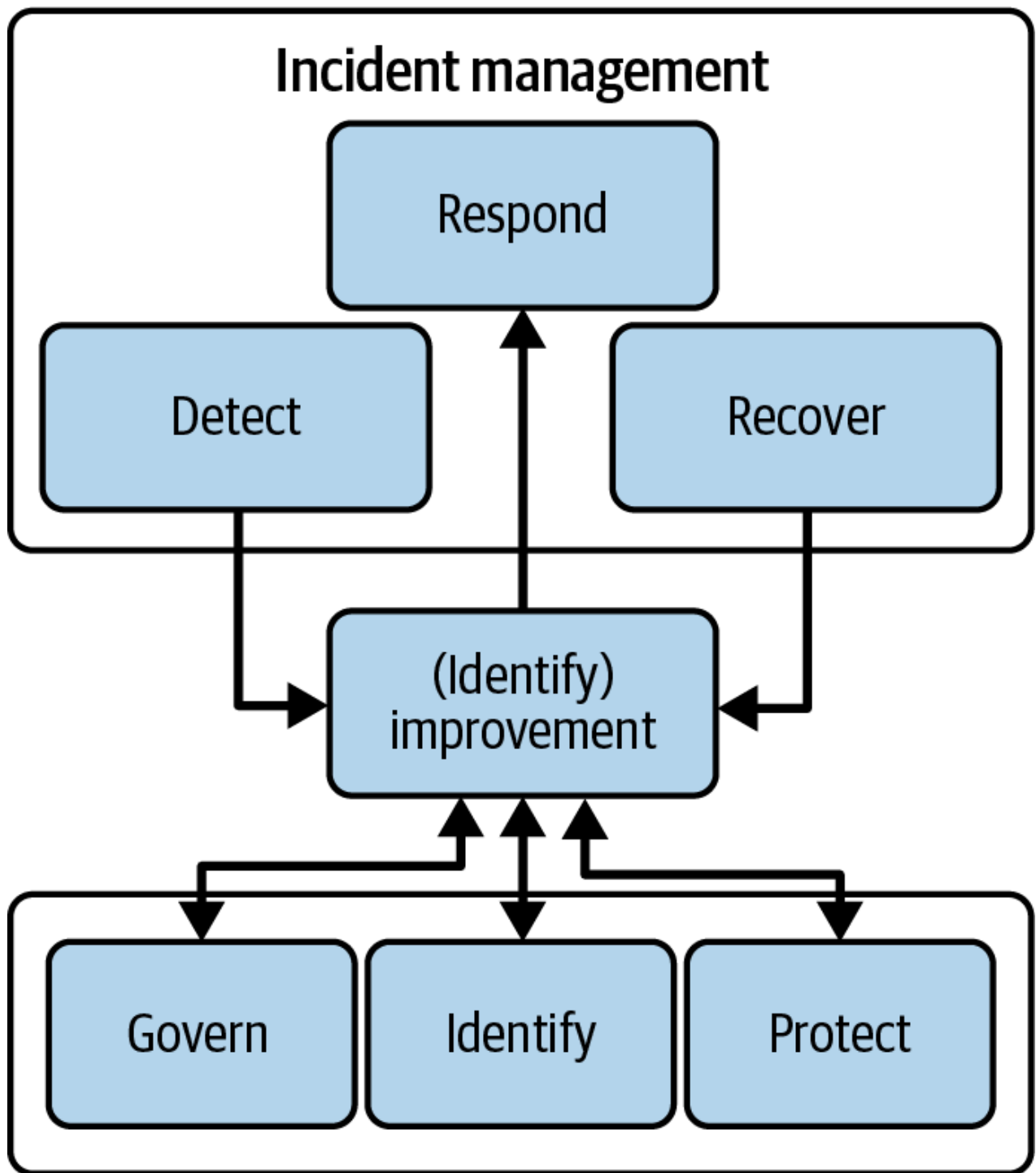
For insight on common weaknesses for web applications and mitigation approaches, consider:

- The **MITRE Common Weakness Enumeration (CWE) system**, which includes over 600 categories across hardware and software
- The **Open Web Application Security Project (OWASP) Top Ten**, which maintains an inventory of the highest impact security risks facing web applications, updated every few years based on trends and impacts

## The NIST Incident Response Lifecycle

When NIST published SP 800-61r2, Incident Response Recommendations and Considerations for Cybersecurity Risk Management, in August 2012, IR was broken down into four main steps: Preparation; Detection and Analysis; Containment, Eradication, and Recovery; and Post-Incident Activity. With the publication of **800-61r3** in April 2025, NIST reworked this model by integrating the Cybersecurity Framework (CSF) into the lifecycle and emphasized a key point: continuous improvement across all aspects of cybersecurity is increasingly necessary, and a linear process no longer suffices.

As shown in **Figure 1-1** and defined by SP 800-61r3, the Detect phase focuses on finding evidence of cybersecurity attacks—whether adversarial or otherwise—and ensuring that compromised assets and accounts are identified and analyzed. The Respond phase encompasses the actions taken following detection. Finally, affected assets and operations are restored during the Recover phase.



*Figure 1-1. The NIST incident response lifecycle*

## The SANS Incident Response Lifecycle

The SANS lifecycle, known as *PICERL*, comprises six steps: Preparation, Identification, Containment, Eradication, Recovery, and Lessons Learned. Shown in **Figure 1-2**, this model is widely adopted and regarded as a standard in the cybersecurity profession outside the US government.

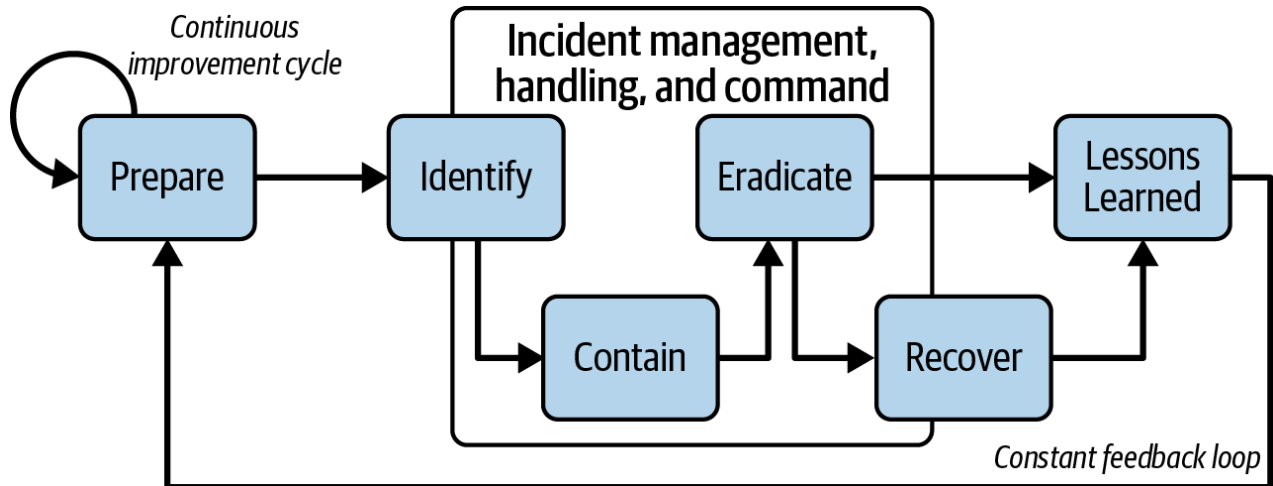


Figure 1-2. The SANS incident response lifecycle

This lifecycle emphasizes that organizations should continuously prepare for incidents by integrating various supporting activities into a feedback loop that forms a continuous cycle throughout and after an incident. As security operations work through alerts, some of the alerts evolve into incidents during the Identification phase when alerts are confirmed not to be false positives or benign activity. Criteria for this verification should be clearly detailed in SecOps playbooks supporting the alert. At this point, depending on the incident's severity and impact, IR and management take over, working through Containment to limit the incident's effects, Eradication to remove the adversary's presence, and Recovery to restore affected systems and operations. Typically, an incident handler is assigned to perform these steps. As the dust settles, the organization engages in the Lessons Learned phase to review the incident and improve its processes, training, and countermeasures.

A particularly strong feature of the PICERL model is its clear language and logical stage gates, which effectively communicate the state and progress throughout the IR lifecycle.

## Dynamic Incident Response and Intelligence Lifecycles

Today, a simple linear model is no longer adequate. As alerts are processed by security operations, threat hunting reveals incidents, or other notifications arise, IR processes and the lifecycles that support them must evolve and become more dynamic. A dynamic model is necessary because incident scope can easily change during the Identification and Containment phases. Modern IR must focus on rapid triage to assess threat levels and facilitate containment decisions, as shown in **Figure 1-3**. As an example, many security operations teams are integrating threat hunting and retroactive threat hunting into their playbooks and processes.

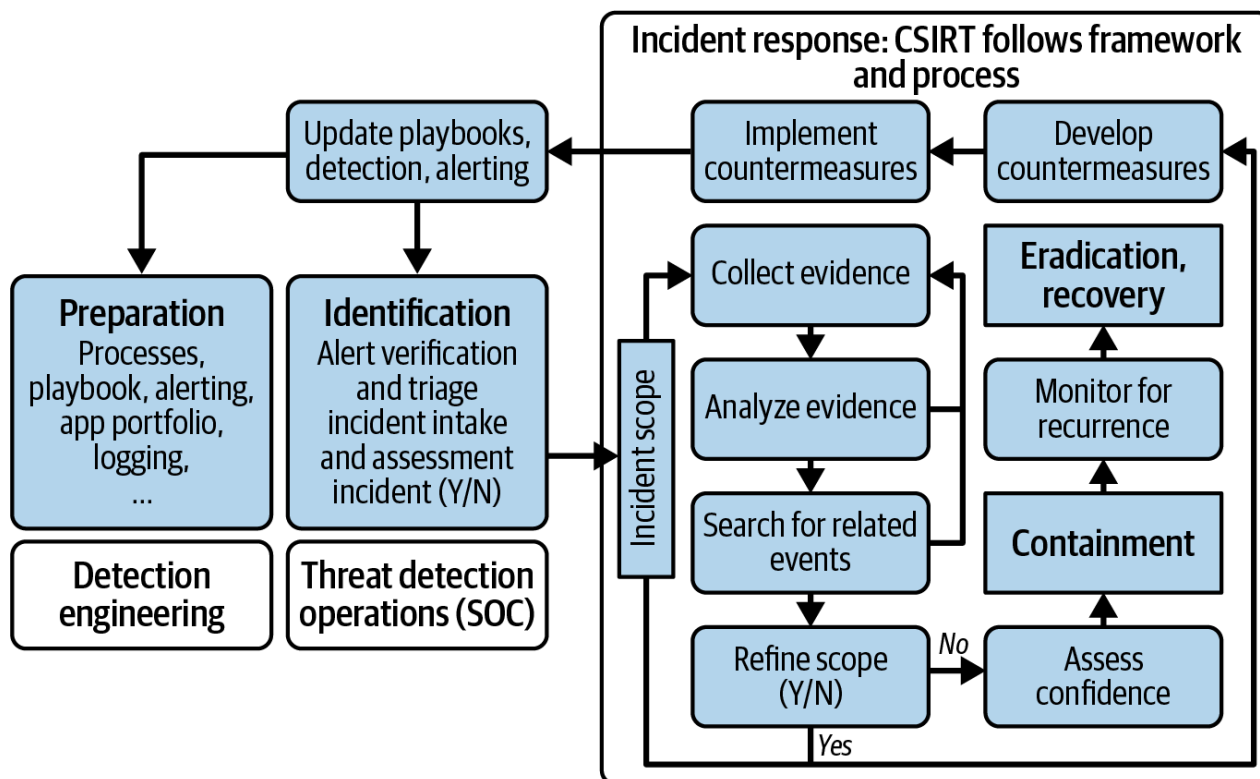


Figure 1-3. Modern dynamic incident response with PICERL

At a practical level, if an indicator—such as a malicious website—is identified, security teams conduct searches (or retroactive hunts) for other users who have visited that website using existing filtering technologies. This can become complicated. For example, a web filter may be an on-premises next-generation firewall (NGFW) or web proxy, or users may utilize different filtering tools when their endpoints are off-site. In some environments, such as cloud-hosted desktops, no web filter may be available. In such cases, surrogate data like current IP addresses and, if available, passive DNS history by day can help perform searches using flow logs. There is often little that can be done for mobile devices such as smartphones, making this a challenging gap to close.

An Endpoint Detection and Response (EDR) tool may alert on hands-on keyboard activity, such as installation of a remote access tool, credential dumping from the LSASS process, creation of a local account added to the Administrators group, or hundreds of other suspicious conditions. At this point, security operations typically escalate the case directly to an incident handler, who will trigger endpoint containment using the EDR tool. This containment action should stop the adversary in their tracks because the endpoint can communicate only with the hosted EDR service as enforced by the local agent. EDR systems provide enterprise-wide capabilities for searching and containment. Meanwhile, adversaries are well armed with advanced techniques and toolkits, enabling domain dominance within a day by a skilled adversary. All an adversary needs is a foothold in a modern network, which they can then leverage to pursue their objectives.

Once an endpoint is contained, it's important to feed the indicators of compromise (IoCs) specific to the case back into the IR process, as illustrated in Figure 1-4, and sweep the enterprise looking for these indicators. Depending on the case, the incident handler may contain more systems and use the IoCs to

search elsewhere in the enterprise for more compromised hosts. Finding indicators on multiple systems may cause a rescoping of the IR process. A small incident may escalate rapidly, requiring staged containment and remediation activities such as changing user credentials and reimaging endpoint systems on staggered schedules. Meanwhile, dedicated teams hunt for “patient zero” by leveraging a wide variety of data sources to uncover the initial intrusion vector. If the incident affects many users and includes users who receive special handling or white glove treatment, those credential resets cannot occur all at once.

Early in the incident, a mini root cause analysis is conducted to understand how the adversary gained access, enabling countermeasures to be developed, tested, and deployed as soon as possible. This usually happens later during the Identification phase or early in Containment.

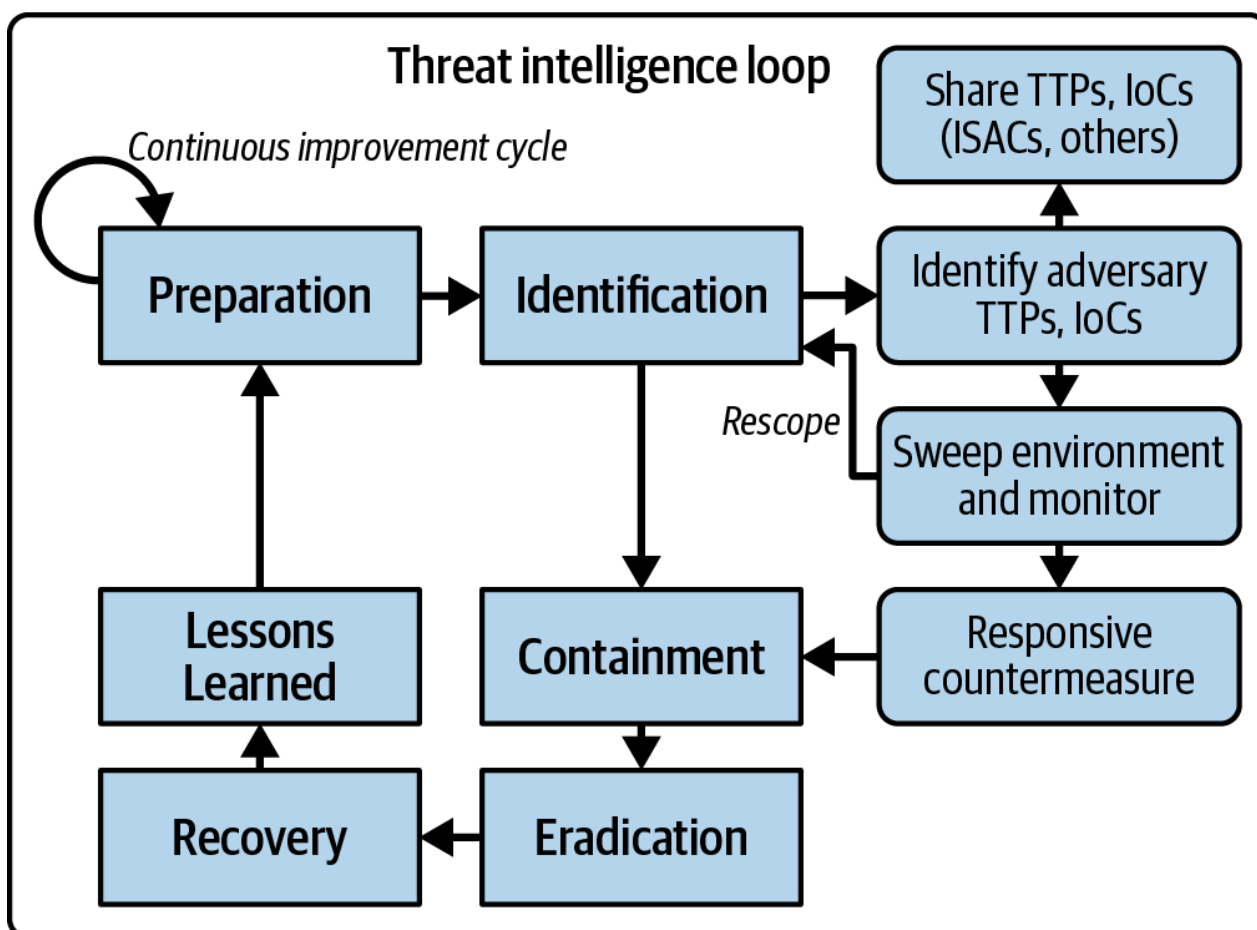


Figure 1-4. Threat intelligence and the IR lifecycle

## Time Based Security

In 1999, Winn Schwartau published *Time Based Security (TBS)*, which centers on measuring a system's security through three time intervals:

*Protection time (Pt)*



This measures how long a system can withstand an attack before becoming compromised. The goal is to implement defenses that increase this time window as much as possible.

#### *Detection time (Dt)*

This measures how long it takes to detect whether an attack is ongoing or has been completed, with the goal of implementing monitoring, intrusion detection, and alerting. Essentially, this includes the time it takes to generate a log record, flow it through a monitoring system such as a SIEM, and have an analyst observe the alert and determine the likelihood that the attack succeeded.

#### *Response time (Rt)*

This measures how long it takes to respond to and mitigate an attack. This time should be minimized via trained staff, IR plans, and as much automation as possible.

Schwartz's formula is simple:

$$Pt > Dt + Rt$$

Even though the model is over 25 years old, it remains relevant today because it provides quantifiable measures that aid in decision making for deploying countermeasures based on cost-effectiveness and time efficiency.

To better understand the practical application of time-based security principles, it is helpful to consider several real-world examples that illustrate the importance of measuring protection, detection, and response times.

In a **notable incident** from March 2025, adversaries exploited a Server-Side Request Forgery (SSRF) attack vector in Jenkins to leak credentials. The detection time was 72 hours, triggered by a budget alarm when adversaries created a malicious p3.8xlarge (8× NVIDIA V100 GPUs) EC2 instance. The overall response time was four hours from alert to key disablement and instance termination. The **financial impact was \$24,000 from a single EC2 instance**. Analysis revealed that the exploitation leveraged two Common Weakness Enumerations (CWEs): **CWE-200**, Exposure of Sensitive Information to an Unauthorized Actor; and **CWE-918**, Server-Side Request Forgery. CWE-200 stemmed from the virtual private cloud (VPC) using Instance Metadata Service IMDSv1, which was triggered because the site allowed SSRF and did not use the more secure IMDSv2. IMDSv2 requires session tokens and additional protections to defend against SSRF attacks.

As a low-cost countermeasure, organizations could implement a CloudTrail monitor to detect p3 instance creation, automatically shut down unauthorized instances with a Lambda function, and notify the monitoring team, unless such instances are being legitimately used. CloudTrail typically delivers log records to S3 within five minutes, enabling response times measured in minutes.

Credential leaks are rapidly weaponized at cloud scale. Palo Alto Networks' Unit 42 released information about the EleKtra-Leak **campaign**, describing how adversaries scanned GitHub repositories

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and leveraged leaked AWS keys within minutes. Although AWS detects and auto-remediates most of these threats today, the protection time for exposed credentials is often measured in minutes—from exposure to malicious use. In other words, exposed credentials can quickly defeat most protection mechanisms.

Effective countermeasures include key rotation, enforcing least-privilege IAM policies, automated secrets scanning in CI/CD pipelines (and related tools such as Jira and Confluence), GitHub Actions secrets scanning, and pattern recognition within data loss prevention (DLP) systems.

## Leveraging MITRE ATT&CK for Incident Response

The MITRE ATT&CK framework, released in May 2015 and frequently updated, is one of the most influential frameworks shaping the IR process. MITRE describes its framework as “a curated knowledge base and model for cyber adversary behavior, reflecting the various phases of an adversary’s attack lifecycle and the platforms they are known to target.” The acronym *ATT&CK* stands for *Adversarial Tactics, Techniques, and Common Knowledge*.

ATT&CK provides a common taxonomy to describe adversary behavior:

- Tactics represent the “why”—the adversary’s goals or objectives.
- Techniques reflect the “how”—the methods the adversary uses to achieve their objective.

By understanding an attacker’s tactics, techniques, and procedures (TTPs), IR moves beyond simply tracking IoCs to understanding attacker behaviors.

Organizations can leverage the framework for cyber defense, security operations, detection engineering, security use case development, and IR to understand adversary behaviors, map techniques and tactics to defensive security architecture, and build better and more actionable detections by applying real-world observations to the security stack. Each technique is assigned a number and subnumber, enabling you to “connect the dots” across systems and know where you are in an attack sequence.

There are several ATT&CK matrices. The main ones are Enterprise, Mobile, Cloud, and Industrial Control Systems. There is also a cloud-specific matrix under the Enterprise domain. Each matrix outlines domain-specific tactics and techniques. The attack process or flow described by a matrix generally moves from left to right. It is critical to understand that the organization or flow is not prescriptive. There is also the downloadable ATT&CK Navigator tool available on [GitHub](#).

The **Enterprise Matrix** covers the following tactics (a partial example is shown in **Figure 1-5**):

- Reconnaissance
- Resource Development

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery

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